

Marine Antifouling Surface Coatings Using Tannic Acid and Poly(*N*-vinylpyrrolidone)

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Development of marine antifouling surface coatings has gained a great deal of attention for the prevention of serious problems such as increased fuel consumption by marine vessels and uncontrolled translocation of marine species.¹ During the last few decades, various approaches were investigated for inhibiting marine fouling of solid substrates,² and biocide-containing paints were successfully utilized for this purpose.³ However, biocide-containing paints were revealed to have adverse effects on the marine ecosystem, and their use was banned by the International Maritime Organization.⁴ Since then, much effort has been devoted for the development of environmentally friendly marine antifouling coatings, *e.g.*, environmentally friendly antifouling polymers have been prepared and used for marine antifouling applications.^{5–17} Statz et al. synthesized poly(ethylene glycol) (PEG) compounds that adhere to titanium surfaces and showed that the PEG-coated surfaces have excellent marine antifouling properties.⁵ Zwitterionic polymers have also been used for marine antifouling coatings.^{10–12} Zhang et al. synthesized polysulfobetaine-coated surfaces on which ultralow marine fouling occurs using surface-initiated polymerization.¹⁰ In addition to the abovementioned polymers, poly(*N*-vinylpyrrolidone) (PVP) has attracted interest as a coating material due to its good antifouling properties and biocompatibility and has been approved by the Food and Drug Administration.^{18–20} However, marine antifouling applications of PVP have not been investigated, because of the lack of a versatile coating method, which is in critical need of being developed.

Recently, the Caruso group reported that the surface of any material can be easily modified using natural polyphenol tannic acid (TA).²¹ Mixing of aqueous solutions of TA and Fe^{III} resulted in the rapid formation of TA–Fe^{III} complexes that were able to coat a wide range of materials, regardless of composition, size, and shape. Moreover, it has been found that abundant hydroxyl groups in the TA enabled facile the formation of hydrogen bonds (HBs) with HB-acceptor polymers.⁹ As PVP is a HB-acceptor polymer, we speculated that a TA coating would serve as an efficient platform for introducing PVP onto the surface of solid substrates.

In this study, we report that a PVP-based marine antifouling surface coating is easily formed on a TA-treated surface *via* multiple HBs. The marine antifouling efficiency of the surface was evaluated using a marine diatom, *Amphora coffeaeformis*, and the TA-mediated PVP coating was found to be highly resistant to diatom adhesion. For all experiments, stainless steel (SS) was used as a model substrate, as it is the major component of marine vessels. Surface coating on SS was carried out by a two-step procedure: polydopamine (pDA)-assisted TA coating²² and PVP coating, resulting in pDA/TA- and pDA/TA/PVP-coated SS (Figure 1). In the first step, the SS surface was pre-treated with pDA coating, which introduces catechol moieties onto the surface.²³ TA coating was then performed on the surface by an Fe^{III} coordination reaction between the catechol moieties of pDA and the pyrogallol moieties of TA.²² Subsequently, PVP coating was carried out by the immersion of pDA/TA-coated surfaces in a PVP solution for 24 h.

The SS surfaces functionalized with pDA/TA and PVP were characterized by water contact angle goniometry and X-ray photoelectron spectroscopy (XPS). The water contact angle of the SS surface decreased from 65.4° to 28.0° and then to 24.8° after pDA/TA and PVP coatings, indicating that the surface wettability of SS was successfully changed (Figure 2(a)).¹⁹ The hydrophilic nature of pDA/TA and PVP led to enhanced surface wettability, as demonstrated by a decrease in the water contact angle. The results of XPS analysis also confirmed the successful introduction of PVP on the SS surface (Figure 2(b)). In the X-ray photoelectron spectrum of the pDA/TA-coated surface, a new N 1s peak of pDA was apparent,²³ and the SS substrate peaks for Cr 2p and Mn 2p disappeared. Calculation of the C/O ratio also supported the successful pDA/TA coating on the surface. The C/O ratio for the SS surface changed from 0.88 to 1.81 after the pDA/TA coating, in agreement with the theoretical ratio of TA (C/O = 1.65). After PVP coating, an increase in the N 1s peak intensity was observed (3.64 → 7.71%), resulting from the high nitrogen content in PVP. All of these observations indicated that the pDA/TA and PVP coatings occurred successfully on the SS surface.

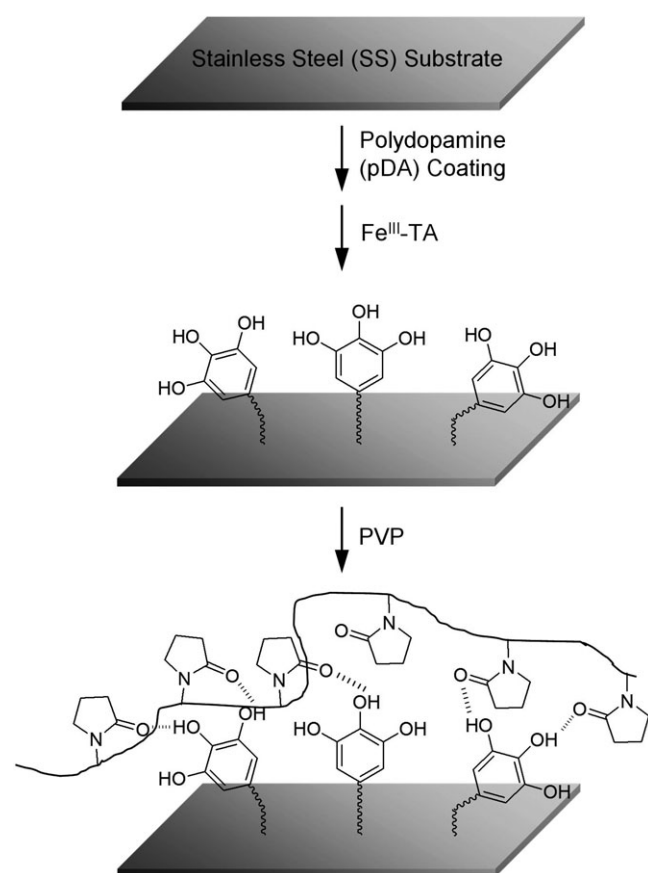


Figure 1. Schematic representation of the formation of marine anti-fouling surface coating with TA and PVP.

To evaluate the marine antifouling performance of the coated surfaces, *A. coffeaeformis* was chosen as a marine fouling organism, as it is commonly found in biofilms on submerged surfaces.²⁴ *A. coffeaeformis* was cultured in an f/2 medium for 10 days, and the SS surfaces were incubated in the diatom suspension (concentration of *chlorophyll a* = 2.0 $\mu\text{g/mL}$) at room temperature. Figure 3 shows the fluorescence images of diatoms attached to differently coated SS surfaces. Incubation of pDA/TA-coated SS in the diatom suspension for 4 h resulted in increased diatom adhesion. Quantitatively, the attached diatom density increased from 57 to 74 diatoms/unit area (unit area: 0.11 mm^2). This increase in attached diatom density is due to the reactivity of TA with proteins.^{25,26} Diatoms are known to use protein-containing glues for adhesion²⁷; therefore, surface deposition of TA on solid substrates can provide a favorable environment for diatom adhesion. This is consistent with our previous studies that showed TA can enhance the initial adhesion of diatoms on solid substrates.^{28,29} Unlike the pDA/TA-coated surfaces, significantly fewer diatoms were able to attach to the pDA/TA/PVP-coated SS. The attached diatom density was 11 diatoms/unit area (unit area: 0.11 mm^2), indicating a fivefold decrease in diatom density compared with that for non-treated SS. The excellent resistance against diatom adhesion originates from the chemical characteristics of PVP. PVP is a

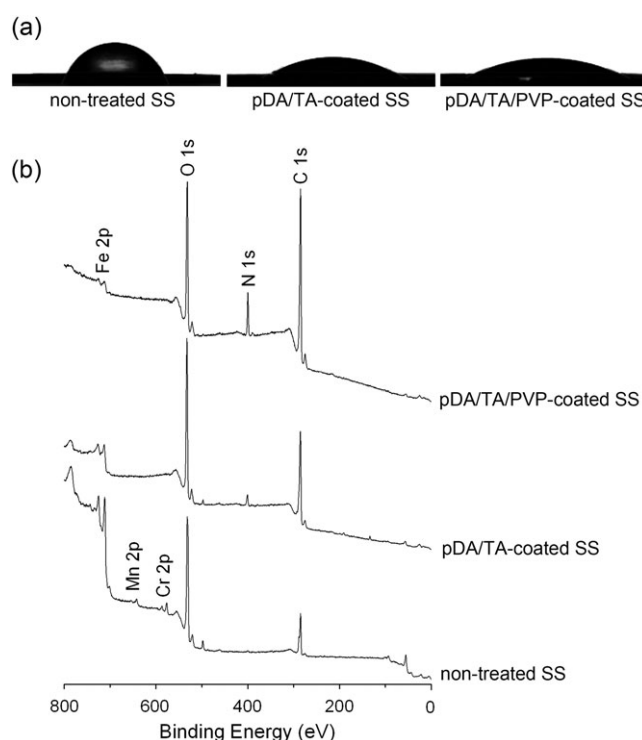


Figure 2. (a) Water contact angles and (b) X-ray photoelectron spectra of different coatings.

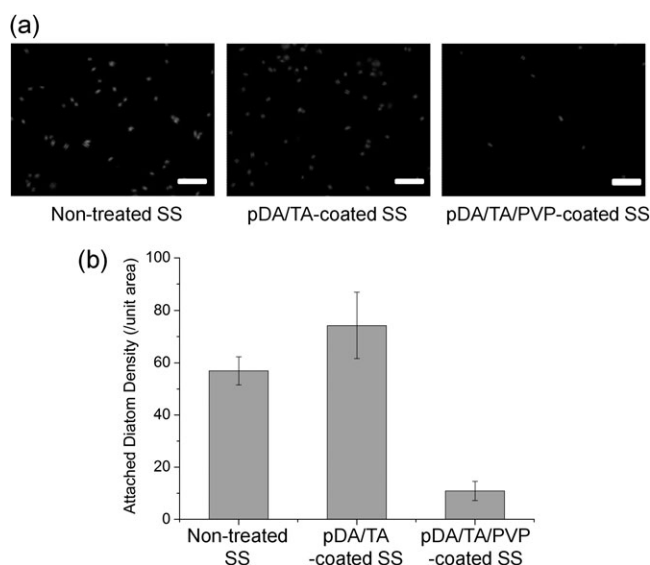


Figure 3. (a) Fluorescence images and (b) quantifications of diatoms attached on the different coatings (unit area: 0.11 mm^2). All scale bars are 50 μm . Each point indicates the mean from 60 counts on three replicate samples, and the error bars display 95% confidence limits.

hydrophilic polymer with many HB accepting groups, no HB donating groups, and no net charge.³⁰ Using the characteristics, PVP lowers the interfacial energy between the surface of the solid substrate and water (*i.e.*, surface hydration). The formation of hydration layers around PVP makes the biofouling onto the surface unfavorable, and the adhesion of microorganisms is inhibited.³⁰

Note

In summary, we developed a facile method for the preparation of marine antifouling surface coatings using polyphenol TA and PVP. TA coating, followed by HB formation between TA and PVP, enabled PVP coating on an SS surface. The resulting coated surface was then used for marine antifouling studies. Incubation of the PVP-coated substrates in a marine diatom suspension presented a significant decrease in the attached diatom density, implying that the marine antifouling properties resulted from the coating. Although HBs between TA and PVP are considered to be unstable on long-term exposure to seawater,⁹ we believe that this method can be a versatile tool for reducing marine fouling on solid substrates, by virtue of the TA coating.

Experimental

Materials. Dopamine hydrochloride (98%, Aldrich, St. Louis, MO, USA), Trizma base (99%, Sigma, Sweden), Trizma HCl (99%, Sigma, St. Louis, MO, USA), iron(III) chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, 97%, Sigma-Aldrich, St. Louis, MO, USA), TA (Sigma-Aldrich, St. Louis, MO, USA), poly(*N*-vinyl pyrrolidone) (average mol wt. 40 000, Sigma-Aldrich, St. Louis, MO, USA), SS ($\text{Fe/Cr}_{18}/\text{Ni}_{10}/\text{Mo}_3$, thickness: 0.05 mm, Goodfellow, Huntingdon, UK), hydrochloric acid (HCl, Tokyo, Junsei), absolute ethanol (Merck, St. Louis, MO, USA), and acetone (99%, Daejung Chemicals & Metals, Siheung City, Korea) were used as received.

pDA and TA Coatings. SS substrates (1 cm $\times$ 1 cm) were cleaned in acetone or ethanol with sonication prior to use. pDA coating was carried out by immersing substrates in a buffer solution (2 mg of dopamine hydrochloride per mL of 10 mM Tris, pH 8.5) at room temperature for 1 h. The coated substrates were rinsed with deionized water and dried under a stream of nitrogen gas. The pDA-coated substrates were transferred to ethanolic solutions of FeCl_3 (10 mM) and TA (10 mM). The following cycle was used for TA coating: (1) FeCl_3 for 1 s, (2) rinsing with ethanol, (3) TA for 1 s, and (4) rinsing with ethanol. Further experiments were carried out after five repetitive treatments with FeCl_3 and TA.

PVP Coating. PVP was grafted to the pDA/TA-coated surfaces by incubating the solid substrates in an acidic solution of PVP (2 mg/mL, 0.01 M HCl) for 24 h. The resulting substrates were rinsed with deionized water and dried under a stream of nitrogen gas.

Diatom Adhesion. *A. coffeaeformis* was received from the Korea Marine Microalgae Culture Center and grown in 100 mL of f/2 culture medium at 18 °C for 10 days. The diatom suspension was added to Petri dishes containing the solid substrates, and after 4 h at room temperature, the substrates were washed in seawater to remove cells that had not attached to the solid substrates. The attached cells were characterized and counted by an optical microscope (Nikon, LV100ND, Tokyo, Japan) equipped with fluorescence imaging system.

Characterization. XPS was carried out using a MultiLab 2000 (Thermo VG Scientific, Waltham, MA, USA) with a Mg K α X-ray source and ultrahigh vacuum ($\sim 10^{-10}$ mbar).

Static water contact angle measurements were carried out using a Phoenix-300 TOUCH goniometer (Surface Electro Optics Co., Ltd., Suwon, Korea).

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